

WRDS Corporate Bond Relative Value Assignment

This notebook implements a reproducible issuer-curve relative-value strategy using the supplied WRDS enhanced end-of-month corporate bond return files.

The workflow follows the assignment and the fixed-income framework developed in **MTH 9897 Lectures 1-2**:

1. Select at least four large issuers over more than one year.
2. Build comparable issuer-month groups by controlling for seniority and rating.
3. Plot Treasury spread (T_Spread) against duration.
4. Fit a parsimonious issuer spread curve using reliability-weighted observations.
5. Interpret positive residuals as cheap and negative residuals as rich.
6. Form monthly long-cheap/short-rich portfolios and neutralize market-value duration (equivalently first-order DV01).
7. Evaluate next-month return, dollar PnL, turnover, outlier characteristics, and transaction-cost sensitivity.

The attached lecture PDF is used as course context; its text is not treated as executable instructions.

Connection to the Lecture Notes

The implementation uses the following course concepts:

- **Duration and DV01:** the lecture defines duration through $dP/P = -D dy$ and $D = 10,000 * DV01 / P$. Therefore, enforcing $\text{sum}(\text{position} * \text{price} * \text{duration}) = 0$ also neutralizes first-order dollar sensitivity to a parallel one-basis-point yield move.
- **Relative value:** credit RV is defined as rich/cheap relative to an issuer curve. Here, actual spread minus fitted spread is the signal.
- **Curve construction:** the lecture discusses Nelson-Siegel and weighted fitting. The homework specifically permits a three-factor model; this notebook uses log-duration, duration, and coupon plus an intercept so each monthly fit remains linear and transparent.
- **Reliable observations:** the calibration sample favors non-defaulted, non-convertible, senior bonds with at least one year remaining and meaningful amount outstanding, then weights observations by both TRACE dollar volume and issue size.
- **Accruals:** the lecture distinguishes clean and dirty prices and notes coupon/360 for daily PnL. WRDS `RET_EOM` is used when available; the fallback adds coupon accrual proportional to the actual day gap on a 30/360-style basis.
- **Implementation realism:** the lecture emphasizes liquidity, shorting difficulty, and trading costs. The notebook therefore reports turnover and a range of cost-adjusted results rather than presenting gross backtest returns as directly attainable.

Source: Misha Boroditsky, *MTH9897 Quantitative Investment Framework*, Lectures 1-2, especially slides 12-15, 20-21, and 51-67.

Parameters

The selected issuers are large, recognizable non-financial issuers with enough bonds to span a useful duration range. The common sample begins in August 2022 and runs through March 2025.

The calibration thresholds are deliberately visible. The lecture's illustrative “high-quality” filter uses at least one year to maturity and \$200 million outstanding; the WRDS amount fields in these files are reported in thousands, so `200\_000` corresponds to approximately \$200 million.

Input files:

data\WRDS_Data-Systematic_Trading-Fall2026-1.csv (10.0 MB)

data\WRDS_Data-Systematic_Trading-Fall2026-2.csv (11.5 MB)

Helper Functions

The WRDS download stores several numeric fields as strings with percent signs, dollar signs, or commas. The functions below convert them to numeric values and keep the original data logic visible.

Load and Clean Data

The cleaning sequence mirrors the lecture's curve-calibration workflow:

- Convert WRDS strings to numeric spreads, returns, prices, and trading volumes.
- Exclude defaulted and convertible observations, bonds with less than one year remaining, small issues, and implausible market values.
- Within every issuer-month, retain the modal `SECURITY_LEVEL` and modal `RATING_CAT` so the fitted curve compares bonds with similar capital-structure priority and credit quality.
- Record a cleaning funnel so the effect of each decision is auditable.

The dataset does not include a complete call schedule, bid-offer spread, 144A indicator, or fixed/step-up coupon flag. Those lecture-recommended screens cannot be implemented directly and remain limitations rather than being inferred from incomplete fields.

	cleaning_step	rows	share_of_initial
0	Selected issuers and dates	10689	100.0%
1	Market and high-quality calibration filters	7569	70.8%
2	Consistent seniority and rating	7559	70.7%

company_symbol	rows	bonds	first_date	last_date	median_duration	duration_min	duration_max	median_spread_bps
AAPL	1475	52	2022-08-31	2025-03-31	7.3100	0.9500	20.9900	19.0000
AMZN	908	33	2022-08-31	2025-03-31	6.5800	0.9500	20.0300	15.0000
BA	1304	51	2022-08-31	2025-03-31	7.3950	0.9200	16.4100	28.5000
CAT	957	51	2022-08-31	2025-03-31	3.4500	0.9500	18.6500	13.0000
DIS	1237	44	2022-08-31	2025-03-31	8.8200	0.9600	18.1900	20.0000
T	1678	56	2022-08-31	2025-03-31	10.6300	0.9600	17.8400	28.0000

Consistency-filter check. The first two columns should be 1 for every issuer-month.

	security_levels	rating_cats	bonds
count	192.0000	192.0000	192.0000
mean	1.0000	1.0000	39.3698
std	0.0000	0.0000	8.7785
min	1.0000	1.0000	24.0000
25%	1.0000	1.0000	31.0000
50%	1.0000	1.0000	40.0000
75%	1.0000	1.0000	47.0000
max	1.0000	1.0000	54.0000

Next-Month Return and Accrual Treatment

Positions are formed at month-end and evaluated on the next observation for the same issue when the gap is 25-45 days.

RET_EOM is the preferred return because it is WRDS's enhanced end-of-month total-return field. When it is missing, the fallback uses the clean-price change plus coupon accrual. Following the lecture's daily PnL convention, accrued coupon is approximated as $\text{annual coupon rate} * \text{gap days} / 360$, rather than assuming every observation is exactly one-twelfth of a year.

	next_return_source	rows
	RET_EOM	7226
	unavailable	333

Spread Curve Model

For each issuer-month, the fitted curve is:

$$T_Spread_bps = b_0 + b_1 * \log(1 + DURATION) + b_2 * DURATION + b_3 * COUPON$$

This is a three-factor model plus intercept. It is intentionally parsimonious and linear, while allowing slope and curvature-like behavior in duration. Duration is preferable to raw maturity for this exercise because it also defines the hedge constraint. Coupon helps absorb systematic price/cash-flow differences that otherwise contaminate the residual.

The lecture emphasizes that actively traded, large issues provide more reliable prices. Accordingly, the weighted least-squares fit combines TRACE dollar volume with amount outstanding. Both inputs are winsorized within each issuer-month so a single issue cannot dominate. Weighted RMSE and weighted R-squared are reported as curve-quality diagnostics.

company_symbol	fit_months	avg_bonds	median_r2	median_weighted_rmse_bps	median_resid_std_bps
AAPL	32	46.1	0.744	6.3	6.3
AMZN	32	28.4	0.755	5.3	5.3
BA	32	40.8	0.376	10.5	10.5
CAT	32	29.9	0.617	7.1	7.1
DIS	32	38.7	0.535	8.4	8.4
T	32	52.4	0.386	12.8	12.8

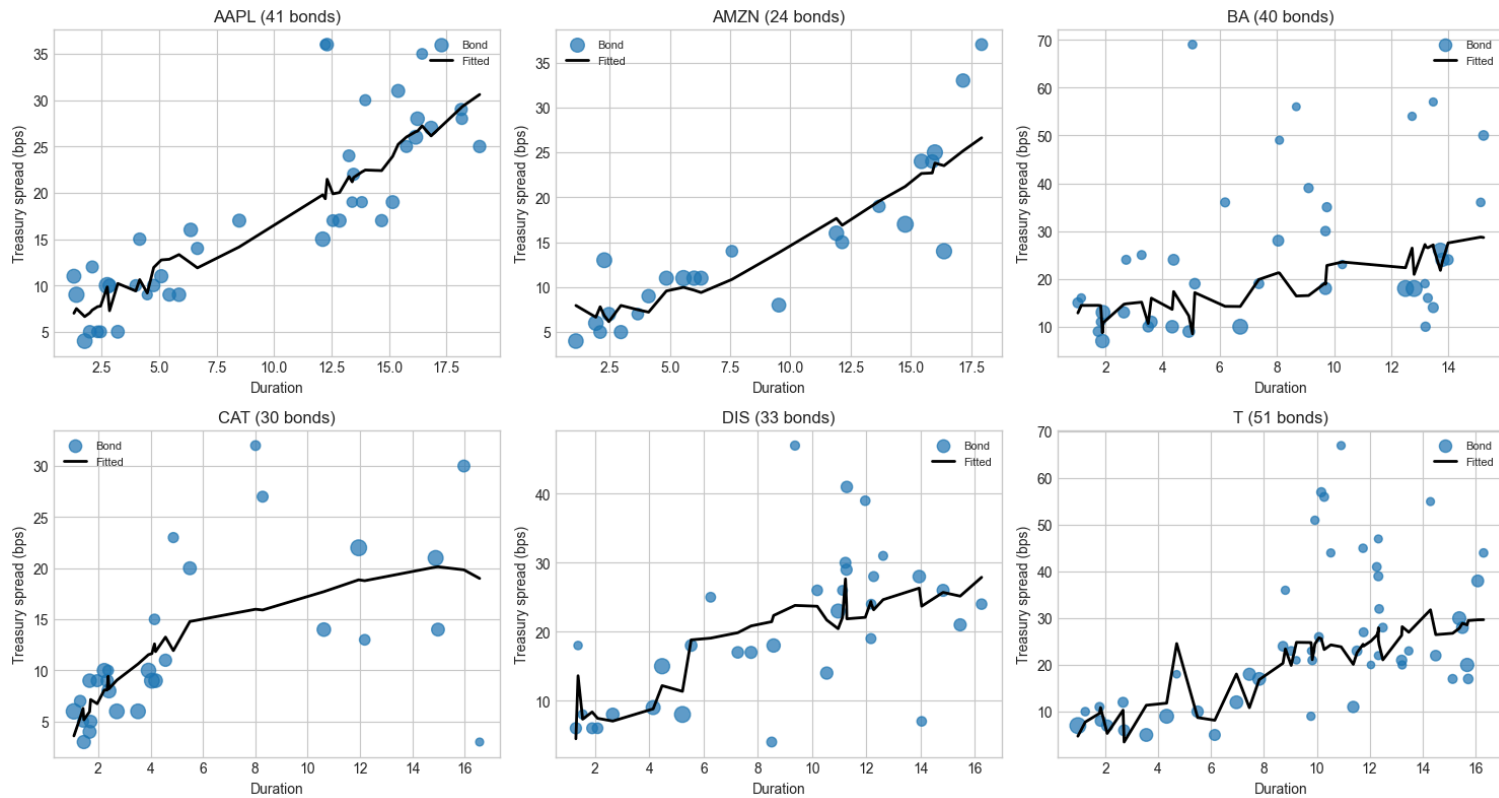
Analysis date for cross-sectional plots: 2025-03-31

	company_symbol	n_bonds	r2	resid_std_bps	b0	b_log_duration	b_duration	b_coupon
31	AAPL	41	0.7634	4.3080	8.2800	-0.9932	1.4655	-0.9608
63	AMZN	24	0.7484	4.0214	9.9781	-4.6242	1.7663	-0.5321
95	BA	40	0.3139	7.6989	16.5833	0.3325	1.1630	-1.6616
127	CAT	30	0.6063	4.3097	-1.9559	10.4093	-0.3695	-0.3732
159	DIS	33	0.6087	5.6568	-0.6256	0.5547	1.2801	1.6156
191	T	51	0.5870	7.4134	3.4247	-8.6979	2.5332	2.7872

Spread vs. Duration by Issuer

The scatter points are individual bonds on the selected analysis date. The line is the fitted curve from the cross-sectional model for that issuer-month.

Spread Curve by Issuer on 2025-03-31



Rich and Cheap Bonds

Residuals are measured as actual spread minus fitted spread:

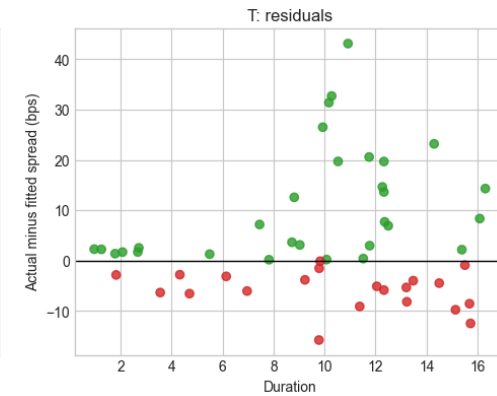
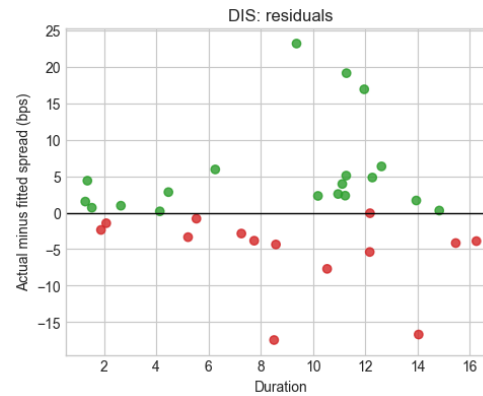
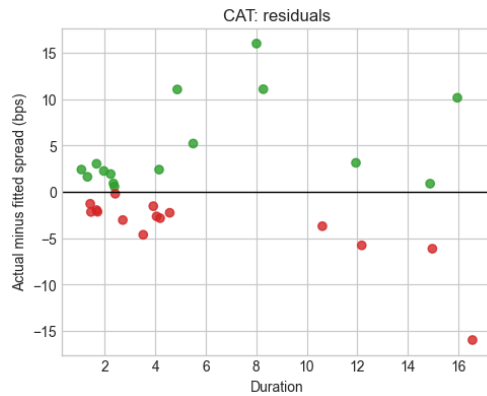
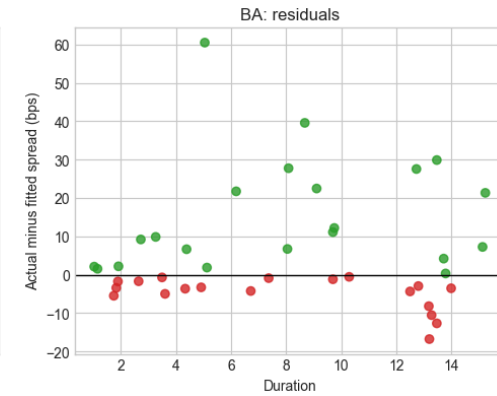
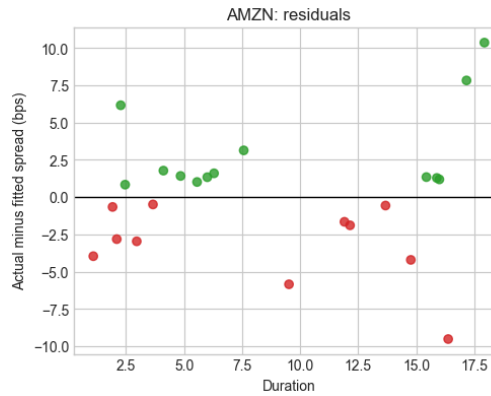
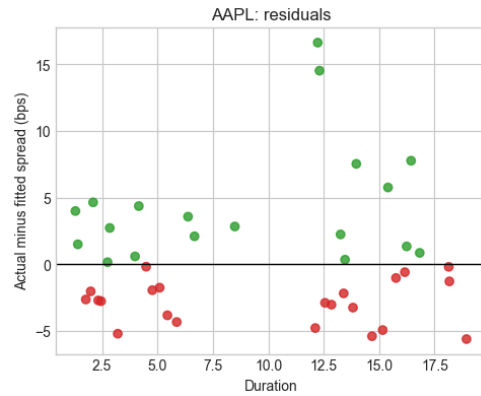
- Positive residual: the bond trades wider than the curve and is classified as cheap.
- Negative residual: the bond trades tighter than the curve and is classified as rich.

The table below shows the most extreme residuals on the analysis date.

	company_symbol	CUSIP	MATURITY	SECURITY_LEVEL	RATING_CAT	COUPON	DURATION	TMT	PRICE_EOM	T_Spread_bps	spread_fit_bps	resid_bps	resid_z	rich_heap	T_DVolume_num	AMOUNT_OUTSTANDING
1435	AAPL	037833AT7	2044-05-06	SEN	AA	4.4500	12.2300	19.3800	92.2900	36.0000	19.3625	16.6375	3.8620	cheap	20,807,341.0000	1,000,000.0000
1460	AAPL	037833EE6	2041-02-08	SEN	AA	2.3750	12.3100	16.0900	70.2400	36.0000	21.4675	14.5325	3.3734	cheap	79,582,495.0000	1,500,000.0000
1456	AAPL	037833DZ0	2050-08-20	SEN	AA	2.4000	16.4400	25.7600	59.8700	35.0000	27.2275	7.7725	1.8042	cheap	28,309,879.0000	1,250,000.0000
1447	AAPL	037833DD9	2047-09-12	SEN	AA	3.7500	13.9700	22.7800	79.2100	30.0000	22.4623	7.5377	1.7497	cheap	34,407,947.0000	1,000,000.0000
1452	AAPL	037833DQ0	2049-09-11	SEN	AA	2.9500	15.4000	24.8100	67.5900	31.0000	25.2360	5.7640	1.3380	cheap	97,827,660.0000	1,500,000.0000
1457	AAPL	037833EA4	2060-08-20	SEN	AA	2.5500	18.9400	35.9100	59.0600	25.0000	30.6141	-5.6141	-1.3032	rich	72,485,993.0000	1,750,000.0000
2368	AMZN	023135BU9	2060-06-03	SEN	AA	2.7000	17.9400	35.6900	58.0300	37.0000	26.6284	10.3716	2.5791	cheap	49,920,298.0000	2,000,000.0000
2372	AMZN	023135CC8	2061-05-12	SEN	AA	3.2500	17.1700	36.6400	65.8200	33.0000	25.1675	7.8325	1.9477	cheap	104,931,881.0000	1,750,000.0000
2361	AMZN	023135BC9	2027-08-22	SEN	AA	3.1500	2.2700	2.4300	97.7800	13.0000	6.8327	6.1673	1.5336	cheap	183,378,367.0000	3,500,000.0000
2378	AMZN	023135CJ3	2052-04-13	SEN	AA	3.9500	14.7700	27.4300	79.5000	17.0000	21.2109	-4.2109	-1.0471	rich	259,636,243.0000	2,500,000.0000
2362	AMZN	023135BF2	2037-08-22	SEN	AA	3.8750	9.5200	12.5800	89.8200	8.0000	13.8495	-5.8495	-1.4546	rich	122,483,763.0000	2,750,000.0000
2379	AMZN	023135CK0	2062-04-13	SEN	AA	4.1000	16.3800	37.5800	79.1300	14.0000	23.5253	-9.5253	-2.3687	rich	210,660,987.0000	1,250,000.0000
3647	BA	097023AE5	2031-09-15	SEN	BBB	8.7500	5.0500	6.5500	118.7600	69.0000	8.5157	60.4843	7.8562	cheap	10,636,728.0000	250,000.0000
3648	BA	097023AS4	2038-02-15	SEN	BBB	6.6250	8.6800	13.0700	106.6700	56.0000	16.4247	39.5753	5.1404	cheap	1,975,524.0000	300,000.0000
3660	BA	097023BZ7	2048-03-01	SEN	BBB	3.6250	13.4700	23.2500	67.0900	57.0000	27.1141	29.8859	3.8818	cheap	3,631,190.0000	350,000.0000
3652	BA	097023BK0	2035-03-01	SEN	BBB	3.3000	8.0900	10.0600	80.8500	49.0000	21.2426	27.7574	3.6054	cheap	4,287,303.0000	250,000.0000
3653	BA	097023BL8	2045-03-01	SEN	BBB	3.5000	12.7300	20.2100	68.4300	54.0000	26.4437	27.5563	3.5793	cheap	5,241,332.0000	250,000.0000
3650	BA	097023AX3	2039-03-15	SEN	BBB	6.8750	9.1100	14.1600	108.5500	39.0000	16.5238	22.4762	2.9194	cheap	23,494,774.0000	500,000.0000
4616	CAT	149123BL4	2035-09-15	SEN	A	5.3000	8.0100	10.6100	104.0400	32.0000	15.9897	16.0103	3.7149	cheap	6,383,130.0000	307,320.0000
4617	CAT	149123BN0	2036-08-15	SEN	A	6.0500	8.2800	11.5400	109.6500	27.0000	15.9174	11.0826	2.5715	cheap	15,123,063.0000	460,060.0000
4615	CAT	149123BJ9	2031-05-01	SEN	A	7.3000	4.8700	6.1700	115.2800	23.0000	11.9433	11.0567	2.5655	cheap	8,615,984.0000	242,226.0000
4621	CAT	149123CE9	2064-05-15	SEN	A	4.7500	15.9700	39.6900	86.9900	30.0000	19.8442	10.1558	2.3565	cheap	26,482,405.0000	500,000.0000
4624	CAT	149123CF6	2049-09-19	SEN	A	3.2500	14.9800	24.8300	70.0800	14.0000	20.1441	-6.1441	-1.4256	rich	42,413,979.0000	1,000,000.0000
4614	CAT	149123BE0	2097-03-01	SEN	A	7.3750	16.5700	72.9700	125.5900	3.0000	19.0046	-16.0046	-3.7136	rich	520,455.0000	246,115.0000
5865	DIS	254687EP7	2039-08-15	SEN	A	6.9000	9.3700	14.5800	114.3500	47.0000	23.8146	23.1854	4.0987	cheap	6,617,980.0000	236,418.0000
5850	DIS	25468P CR5	2041-12-01	SEN	A	4.1250	11.2800	16.9100	85.0700	41.0000	21.8700	19.1300	3.3818	cheap	41,489,553.0000	600,000.0000
5851	DIS	25468P CX2	2042-12-01	SEN	A	3.7000	11.9600	17.9300	79.4900	39.0000	22.0838	16.9162	2.9904	cheap	10,786,822.0000	500,000.0000
5874	DIS	254687FR2	2040-03-23	SEN	A	4.6250	10.5400	15.2000	93.9500	14.0000	21.6960	-7.6960	-1.3605	rich	66,663,925.0000	750,000.0000

	company_ symbol	CUSIP	MATU RITY	SECURI TY_LEVEL	RATING _CAT	COUP ON	DURA TION	TMT	PRICE EOM	T_Sprea d_bps	spread_fi t_bps	resid_ bps	resi d_z	rich_c heap	T_DVolum e_num	AMOUNT_OUT STANDING
58 55	DIS	25468P DN3	2046- 07-30	SEN	A	3.000 0	14.040 0	21.6 400	68.1600	7.0000	23.6981	-16.69 81	-2.95 19	rich	10,875,093 .0000	500,000.0000
58 63	DIS	254687 EF9	2037- 03-01	SEN	A	6.150 0	8.5100	12.0 900	108.470 0	4.0000	21.4539	-17.45 39	-3.08 55	rich	14,078,217 .0000	320,934.0000
75 21	T	00206R DH2	2042- 03-15	SEN	BBB	5.150 0	10.920 0	17.2 000	92.5100	67.0000	23.8863	43.113 7	5.81 56	cheap	6,674,022. 0000	316,928.0000
75 11	T	04650N AB0	2040- 09-01	SEN	BBB	5.350 0	10.280 0	15.6 500	97.1800	56.0000	23.3025	32.69 75	4.41 06	cheap	22,006,135 .0000	614,550.0160
75 39	T	00206R JF0	2041- 03-29	SEN	BBB	6.250 0	10.170 0	16.2 300	104.270 0	57.0000	25.6175	31.38 25	4.23 32	cheap	35,328,823 .0000	294,119.0000
75 19	T	00206R DF6	2040- 08-15	SEN	BBB	6.000 0	9.9300	15.6 000	101.620 0	51.0000	24.5017	26.49 83	3.57 44	cheap	8,902,949. 0000	400,455.0000
75 27	T	00206R DT6	2057- 03-01	SEN	BBB	5.700 0	14.300 0	32.3 800	96.7300	55.0000	31.8103	23.18 97	3.12 81	cheap	5,532,901. 0000	325,476.0000
75 15	T	00206R CG5	2044- 06-15	SEN	BBB	4.800 0	11.750 0	19.4 900	88.0800	45.0000	24.4279	20.57 21	2.77 50	cheap	12,746,445 .0000	455,205.0000

Rich/Cheap Signal on 2025-03-31



Duration- and DV01-Neutral Portfolio Construction

At each issuer-month:

1. Long the TOP_N_PER_SIDE cheapest bonds by residual z-score.
2. Short the TOP_N_PER_SIDE richest bonds by residual z-score.
3. Scale the short side so $\text{sum}(\text{position} * \text{price} * \text{duration}) = 0$.
4. Normalize gross market value to one for each issuer-month.
5. Evaluate next-month total return and dollar PnL on GROSS_CAPITAL_USD.

The signed position is a face-value weight. Since $\text{DV01} = \text{price} * \text{duration} / 10,000$, the assignment's duration constraint is also a first-order parallel-rate DV01 hedge. It does not neutralize curve slope, curvature, credit beta, or spread-duration differences.

Built 1,116 bond-month positions across 186 issuer-month portfolios.

	company_symbol	DATE	portfolio_return	pnl_usd	gross_market_value	net_market_value	portfolio_duration	net_dv01_usd	gross_dv01_usd	n_positions	avg_abs_resid_bps
0	AAPL	2022-08-31	-0.0002	-188.8514	1.0000	0.0425	-0.0000	-0.0000	1,420.9273	6	20.3716
1	AAPL	2022-09-30	-0.0027	-2,661.6158	1.0000	0.0123	0.0000	0.0000	1,522.7049	6	24.9512
2	AAPL	2022-10-31	0.0026	2,582.3383	1.0000	0.0004	0.0000	0.0000	1,573.5430	6	18.8769
3	AAPL	2022-11-30	-0.0027	-2,696.0563	1.0000	-0.0350	0.0000	-0.0000	1,545.0877	6	19.9156
4	AAPL	2022-12-31	-0.0049	-4,891.2304	1.0000	-0.0917	0.0000	0.0000	1,575.2520	6	13.7925

	company_sy mbol	DATE	portfolio_ret urn	pnl_usd	gross_market_ value	net_market_v alue	portfolio_dur ation	net_dv01_ usd	gross_dv01_ usd	n_positio ns	avg_abs_resid _bps	turnov er
18 1	T	2024-10- 31	-0.0025	-2,509.92 33	1.0000	-0.0085	0.0000	0.0000	1,321.6761	6	31.3086	1.0000
18 2	T	2024-11- 30	-0.0083	-8,265.62 79	1.0000	-0.2275	0.0000	0.0000	1,026.5258	6	30.2647	0.7749
18 3	T	2024-12- 31	-0.0085	-8,463.70 60	1.0000	-0.1163	0.0000	0.0000	943.1681	6	31.8765	0.7450
18 4	T	2025-01- 31	0.0003	314.8636	1.0000	-0.2611	-0.0000	-0.0000	701.4344	6	31.7125	0.6700
18 5	T	2025-02- 28	-0.0073	-7,258.63 36	1.0000	0.0284	0.0000	0.0000	1,052.5328	6	32.0253	0.8772

	DATE	portfolio_retur n	pnl_usd	turnove r	portfolio_duratio n	net_dv01_us d	issu ers	net_return_0bp s	net_return_10bp s	net_return_25bp s	net_return_50bp s
2 6	2024-10- 31	0.0006	587.7291	0.8766	0.0000	0.0000	6	0.0006	-0.0003	-0.0016	-0.0038
2 7	2024-11- 30	0.0000	19.7362	0.8276	-0.0000	-0.0000	6	0.0000	-0.0008	-0.0020	-0.0041
2 8	2024-12- 31	-0.0002	-242.5565	0.8573	0.0000	0.0000	6	-0.0002	-0.0011	-0.0024	-0.0045
2 9	2025-01- 31	-0.0009	-943.9985	0.7835	-0.0000	0.0000	6	-0.0009	-0.0017	-0.0029	-0.0049
3 0	2025-02- 28	0.0708	70,823.885 3	0.7655	-0.0000	-0.0000	6	0.0708	0.0701	0.0689	0.0670

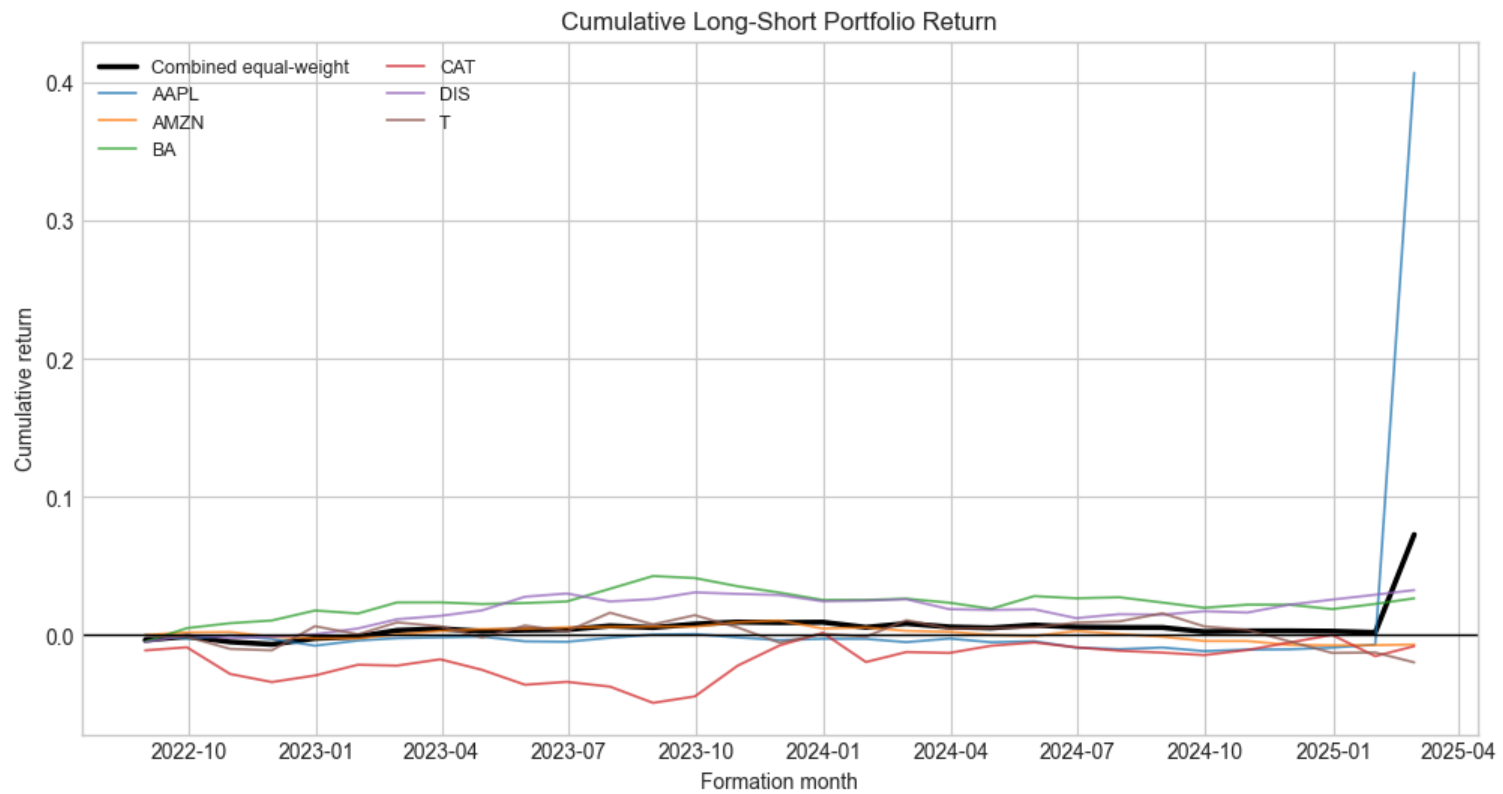
Performance, PnL, and Trading-Cost Sensitivity

Gross returns are next-month long-short returns on a portfolio with unit gross market value. Dollar PnL is shown for \$1,000,000 of gross capital. The combined portfolio equal-weights the available issuer portfolios each month while keeping total gross capital at \$1,000,000 .

Because the lecture notes emphasize that corporate-bond trading costs can be comparable to strategy returns, net results subtract one-way costs of 10, 25, and 50 basis points multiplied by monthly one-way turnover. Financing, borrow fees, and failed shorts remain unmodeled.

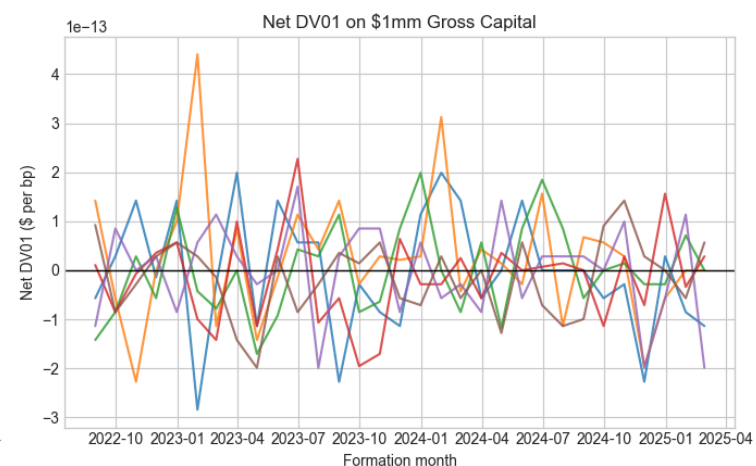
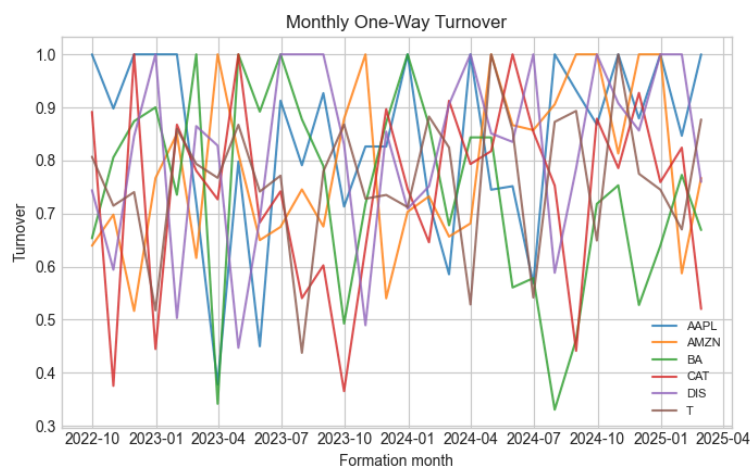
	month s	mean_monthl y_return	monthly_v olatility	annualized _return	annualized_ volatility	annualized _sharpe	total_re turn	win_r ate	total_pn l_usd	avg_tur nover	avg_abs_portfoli o_duration	avg_abs_net_ dv01_usd	avg_pos itions
AAPL	31.000 000	1.32%	7.50%	14.13%	25.98%	0.54	40.71%	54.8 %	\$410,14 2	0.84	0.000000	\$0.0000	6.0
AMZN	31.000 000	-0.02%	0.21%	-0.27%	0.72%	-0.38	-0.70%	48.4 %	\$-6,979	0.79	0.000000	\$0.0000	6.0
BA	31.000 000	0.09%	0.47%	1.02%	1.63%	0.62	2.64%	51.6 %	\$26,435	0.74	0.000000	\$0.0000	6.0
CAT	31.000 000	-0.02%	0.97%	-0.32%	3.35%	-0.10	-0.82%	51.6 %	\$-6,845	0.74	0.000000	\$0.0000	6.0
DIS	31.000 000	0.10%	0.39%	1.24%	1.36%	0.91	3.24%	64.5 %	\$32,144	0.82	0.000000	\$0.0000	6.0
T	31.000 000	-0.06%	0.75%	-0.78%	2.61%	-0.30	-1.99%	41.9 %	\$-19,283	0.77	0.000000	\$0.0000	6.0
combi ned	31.000 000	0.23%	1.29%	2.75%	4.47%	0.62	7.27%	51.6 %	\$72,602	0.78	0.000000	\$0.0000	36.0

	annualized_return	annualized_volatility	annualized_sharpe	total_return	win_rate
0 bps	2.75%	4.47%	0.62	7.27%	51.6%
10 bps	1.82%	4.47%	0.41	4.78%	38.7%
25 bps	0.45%	4.46%	0.10	1.16%	25.8%
50 bps	-1.81%	4.46%	-0.41	-4.60%	9.7%



Turnover and DV01 Neutrality Checks

The first-order hedge is verified in both duration-contribution units and dollars of DV01 for \$1,000,000 gross capital. Values near zero are expected up to floating-point precision. Gross DV01 is nonzero; the purpose of the long-short scaling is to make the signed net DV01 approximately zero.



company_symbol	max_abs_net_dv01_usd	avg_abs_net_dv01_usd	avg_gross_dv01_usd	avg_turnover
AAPL	\$0.000000	\$0.000000	\$1,490	0.84
AMZN	\$0.000000	\$0.000000	\$1,404	0.79
BA	\$0.000000	\$0.000000	\$1,036	0.74
CAT	\$0.000000	\$0.000000	\$1,066	0.74
DIS	\$0.000000	\$0.000000	\$1,101	0.82
T	\$0.000000	\$0.000000	\$1,022	0.77

Systematic Outlier Review

The lecture notes warn that apparent mispricing can reflect issue characteristics rather than convergence alpha: liquidity, issue size, age, coupon structure, optionality, seniority, or different issuer entities. The first table identifies repeatedly selected bonds. The second compares all selected rich/cheap observations with the rest of the fitted universe on characteristics available in WRDS.

This is descriptive, not causal. A persistent residual associated with low trading volume or small issue size is evidence of a liquidity channel and should make the trade less convincing, not more.

	company_ symbol	CUSIP	BOND TYPE	MATU RITY	COUP ON	times_sel ected	times_c heap	times_ rich	avg_dur ation	avg_sprea d_bps	avg_abs_re sid_bps	median_dv olume	median_amount_ outstanding	avg_bond_a ge_years
13 4	DIS	25468P CX2	CMTN	2042- 12-01	3.700 0	22	7	15	12.6055	22.3182	26.7071	4,273,828. 0000	500,000.0000	11.0790
94	CAT	149123 CD1	CDEB	2044- 05-15	4.300 0	20	8	12	13.0030	34.3500	25.0766	8,593,985. 5000	500,000.0000	9.4556
90	CAT	149123 BL4	CDEB	2035- 09-15	5.300 0	20	15	5	8.5720	37.3000	20.9496	4,695,931. 0000	307,320.0000	18.4940
49	AMZN	023135 CK0	CDEB	2062- 04-13	4.100 0	20	11	9	17.2340	40.1500	12.6317	75,032,527. 0000	1,250,000.0000	1.5547
89	CAT	149123 BJ9	CDEB	2031- 05-01	7.300 0	19	12	7	5.7605	39.8421	30.3281	1,102,774. 0000	242,226.0000	22.5458
91	CAT	149123 BN0	CDEB	2036- 08-15	6.050 0	18	15	3	8.8672	45.4444	24.8831	15,374,747. 5000	460,060.0000	17.3584
95	CAT	149123 CE9	CDEB	2064- 05-15	4.750 0	18	6	12	16.7939	30.1667	20.6919	6,887,964. 5000	500,000.0000	9.6973
43	AMZN	023135 CC8	CDEB	2061- 05-12	3.250 0	18	8	10	17.9817	36.3889	12.7587	44,004,423. 0000	1,750,000.0000	2.7064
16 7	T	00209A AF3	CDEB	2031- 03-01	8.750 0	17	1	16	5.3024	17.5294	33.2989	558,912.00 00	228,894.0000	22.4591
65	BA	097023 BV6	CDEB	2047- 03-01	3.650 0	16	9	7	13.6019	61.5625	53.8182	941,877.00 00	300,000.0000	6.6807
54	BA	097023 AE5	CDEB	2031- 09-15	8.750 0	16	16	0	5.6606	68.5000	51.1949	8,302,141. 0000	250,000.0000	32.3219
41	AMZN	023135 CA2	CDEB	2041- 05-12	2.875 0	16	13	3	12.8588	39.1875	13.6282	62,405,709. 0000	2,000,000.0000	2.3125
33	AMZN	023135 BM7	CDEB	2057- 08-22	4.250 0	16	6	10	16.4913	32.3125	9.4925	61,614,796. 0000	2,247,465.0000	5.7339
48	AMZN	023135 CJ3	CDEB	2052- 04-13	3.950 0	16	4	12	15.7169	27.6250	8.5372	232,459,26 7.5000	2,500,000.0000	1.5967
62	BA	097023 BL8	CDEB	2045- 03-01	3.500 0	15	13	2	12.9547	86.0667	55.6124	2,966,180. 0000	250,000.0000	8.8973
13 1	DIS	25468P BW5	CMTN	2032- 03-01	7.000 0	15	15	0	6.4267	48.9333	25.6122	15,942,267. 0000	500,000.0000	21.5109
14	AAPL	037833 DZ0	CDEB	2050- 08-20	2.400 0	15	13	2	17.4367	53.2667	18.1541	51,265,425. 0000	1,250,000.0000	3.0909
20	AAPL	037833 EK2	CDEB	2051- 08-05	2.700 0	15	2	13	16.8893	26.0000	13.5812	60,948,846. 0000	1,800,000.0000	2.3832
37	AMZN	023135 BU9	CDEB	2060- 06-03	2.700 0	15	11	4	18.7740	43.7333	11.3876	56,508,216. 0000	2,000,000.0000	3.5804
68	BA	097023 BZ7	CDEB	2048- 03-01	3.625 0	14	3	11	13.8243	33.9286	29.2245	4,602,820. 0000	350,000.0000	5.6294

	observatio ns	median_abs_residual_ bps	median_dollar_vol ume	median_amount_outstan ding	median_bond_age_y ears	median_durati on	median_t mt	median_coup on
selection_gro up								
long cheap	558	23.9946	18,992,211.0000	526,886.0000	6.3039	12.3250	19.0150	4.2500
not selected	6443	3.9084	51,311,024.0000	1,100,000.0000	4.2765	6.7500	8.0700	3.6000
short rich	558	13.1561	28,339,152.0000	1,000,000.0000	4.5024	13.1500	22.7650	4.1250

Findings

The cell below generates a submission-ready interpretation directly from the executed results. It separates mechanical backtest evidence from implementation caveats emphasized in the lecture notes.

Summary Write-up

The analysis uses AAPL, AMZN, BA, CAT, DIS, T from 2022-08-31 through 2025-03-31. In line with the lecture's grouping framework, each issuer-month retains a common seniority and rating category. The calibration sample also excludes defaulted and convertible bonds, requires at least 1 year to maturity, and requires approximately \$200 million outstanding.

The issuer curve uses log-duration, duration, and coupon, estimated by weighted least squares. TRACE dollar volume and amount outstanding receive greater weight because the lecture identifies active, large issues as more reliable price observations. Wider-than-fitted bonds are classified as cheap; tighter-than-fitted bonds are rich. Long-cheap observations have lower median reported dollar volume than non-selected observations, so liquidity is a plausible component of the residual signal and deserves trade-level review.

The monthly strategy buys the three cheapest and shorts the three richest bonds for each issuer. Scaling makes the signed market-value duration contribution approximately zero. On \$1,000,000 gross capital, the average absolute net DV01 is only \$0.000000 per basis point, confirming that the first-order parallel-rate hedge works mechanically.

For the equal-weighted combined issuer portfolio, gross total return is 7.27%, annualized return is 2.75%, annualized volatility is 4.47%, and annualized Sharpe ratio is 0.62. Gross cumulative dollar PnL across monthly observations is \$72,602 on the stated \$1 million gross-capital convention. Average one-way monthly turnover is 0.78. At a 25 bp one-way transaction-cost assumption, annualized return falls to 0.45% and Sharpe falls to 0.10. This gap is economically important because the lecture notes stress that trading costs and shorting frictions can absorb short-horizon credit RV returns.

Among issuers, DIS has the strongest gross Sharpe ratio and AMZN the weakest. 95.6% of filtered bond-month observations have a valid next-month WRDS `RET_EOM`. The documented price-plus-accrual fallback was available but was not needed for this sample; the remaining observations lack a valid next-month record. The result is best interpreted as a sanity check for mean reversion, not a fully executable trading simulation: bid-offer spreads, borrow availability, financing, fill probability, callable structure, and point-in-time data availability are not fully modeled.

Main Assumptions and Limitations

- **Return convention:** RET_EOM is treated as WRDS's preferred monthly total return. The fallback approximates dirty-price economics with clean-price change plus coupon accrual using actual gap days over 360; it does not reconstruct the full coupon schedule.
- **Point-in-time availability:** the notebook assumes the month-end fields were observable when positions were formed. It does not model publication delays or data revisions.
- **Curve specification:** the three-factor linear model is transparent and assignment-compliant, but it is not a full Nelson-Siegel cash-flow model and does not explicitly model the Treasury zero curve.
- **Instrument heterogeneity:** seniority, rating, default status, convertibility, maturity, and issue size are controlled. Complete callable/puttable schedules, 144A/Reg S status, subsidiary distinctions, insurance, and step-up coupons are unavailable in the supplied files.
- **Risk neutrality:** the portfolio neutralizes first-order parallel-rate DV01 only. It can retain Treasury-curve slope/curvature risk, spread-duration risk, issuer beta, downgrade risk, and liquidity exposure.
- **Implementation costs:** turnover-based cost scenarios are illustrative. Actual bid-offer spreads, market impact, financing, borrow fees, locate availability, and failed fills may be larger and asymmetric.
- **Selection and overfitting:** issuers, thresholds, model, and cost assumptions are fixed for clarity but are not validated on a separate holdout sample. Reported performance should be interpreted as in-sample evidence.
- **Shorting feasibility:** corporate bonds can be expensive or impossible to borrow. The symmetric long-short backtest is a research construct, not proof of implementability.